Project Restructure: Descriptive Analysis

Notes 3 Main areas of analysis:

- Temporal
- Institutional
- Geographic

Did not include all territories and jurisdictions that analysis will be done separately. Focusing solely on the U.S. states.

Temporal Analysis

Question How has undergraduate enrollment changed across U.S. states from 2012-2022, and are these changes statistically significant?

In [111…

```
import numpy as np
import pandas as pd
from scipy import stats
from scipy.stats import t
import plotly.express as px
import plotly.graph_objects as go
import os
import pymannkendall as mk
from plotly.subplots import make_subplots
```

In [112…

```
enrollment_df = pd.read_csv(os.path.dirname(os.getcwd()) + '/data/cleaned_er
enrollment_df.rename(columns={'State or jurisdiction': 'State'}, inplace=Tru
enrollment_df.head()
```

Out[112…

	State	Year	Total_Pub_Under	4y_Pub_Under	2y_Pub_Under	Total_Pub_Postba
0	Alabama	2012-13	216535	130260	86275	345
1	Alabama	2013-14	213669	129045	84624	346
2	Alabama	2014-15	212458	129916	82542	345
3	Alabama	2015-16	212968	131503	81465	344
4	Alabama	2016-17	216115	135080	81035	349

In [113…

```
states = [
    "Alabama", "Alaska", "Arizona", "Arkansas", "California", "Colorado", "C
    "Delaware", "Florida", "Georgia", "Hawaii", "Idaho", "Illinois", "Indiar
```

```
    "Kansas", "Kentucky", "Louisiana", "Maine", "Maryland", "Massachusetts",
    "Minnesota", "Mississippi", "Missouri", "Montana", "Nebraska", "Nevada",
    "New Jersey", "New Mexico", "New York", "North Carolina", "North Dakota"
    "Oklahoma", "Oregon", "Pennsylvania", "Rhode Island", "South Carolina",
    "Tennessee", "Texas", "Utah", "Vermont", "Virginia", "Washington", "West
    "Wisconsin", "Wyoming", "District of Columbia"
]

state_enrollment = enrollment_df[enrollment_df["State"].isin(states)]
state_enrollment['State'].unique()
```

Out[113…

```
array(['Alabama', 'Alaska', 'Arizona', 'Arkansas', 'California',
       'Colorado', 'Connecticut', 'Delaware', 'District of Columbia',
       'Florida', 'Georgia', 'Hawaii', 'Idaho', 'Illinois', 'Indiana',
       'Iowa', 'Kansas', 'Kentucky', 'Louisiana', 'Maine', 'Maryland',
       'Massachusetts', 'Michigan', 'Minnesota', 'Mississippi',
       'Missouri', 'Montana', 'Nebraska', 'Nevada', 'New Hampshire',
       'New Jersey', 'New Mexico', 'New York', 'North Carolina',
       'North Dakota', 'Ohio', 'Oklahoma', 'Oregon', 'Pennsylvania',
       'Rhode Island', 'South Carolina', 'South Dakota', 'Tennessee',
       'Texas', 'Utah', 'Vermont', 'Virginia', 'Washington',
       'West Virginia', 'Wisconsin', 'Wyoming'], dtype=object)
```

Sub-Question What is the national enrollment trend when aggregating across all states?

In [114…

```
state_enrollment['Total_Enrollment'] = state_enrollment['Total_Pub_Under'] +
state_enrollment.head()
```

```
/var/folders/6g/_wpr8bs53rs84lqb6nn_rk900000gn/T/ipykernel_59561/1575148348.
py:1: SettingWithCopyWarning:


A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/
stable/user_guide/indexing.html#returning-a-view-versus-a-copy
```

Out[114…

	State	Year	Total_Pub_Under	4y_Pub_Under	2y_Pub_Under	Total_Pub_Postba
0	Alabama	2012-13	216535	130260	86275	345
1	Alabama	2013-14	213669	129045	84624	346
2	Alabama	2014-15	212458	129916	82542	345
3	Alabama	2015-16	212968	131503	81465	344
4	Alabama	2016-17	216115	135080	81035	349

In [115…

```
nationwide_trends = state_enrollment.groupby('Year').agg({'Total_Enrollment'
nationwide_trends
```

Out[115…

	Year	Total_Enrollment	Total_Pub_Under	Total_Priv_Under
0	2012-13	17717229	13458541	4258688
1	2013-14	17459865	13332032	4127833
2	2014-15	17278075	13230125	4047950
3	2015-16	17021992	13130934	3891058
4	2016-17	16854168	13126042	3728126
5	2017-18	16745071	13085693	3659378
6	2018-19	16594731	13033822	3560909
7	2019-20	16549691	12986168	3563523
8	2020-21	15836385	12305625	3530760
9	2021-22	15433062	11929275	3503787

In [116…

```
baseline_year = nationwide_trends[nationwide_trends['Year'] == '2012-13']
baseline_enrollment = nationwide_trends['Total_Enrollment'][0]
nationwide_trends['enrollment_index'] = (nationwide_trends['Total_Enrollment
nationwide_trends['yoy_change'] = nationwide_trends['Total_Enrollment'].pct_
```

In [117…

```
total_change_pct = ((nationwide_trends['Total_Enrollment'][9] - nationwide_t
print(total_change_pct)
```

```
-12.892349023653754
```

In [118…

```
years = np.arange(len(nationwide_trends))
slope,intercept, r_value, p_value, std_err = stats.linregress(years, nationw
```

In [119…

```
n = len(years)
dof = n - 2
t_crit = t.ppf(0.975, dof)  # 95% CI
ci_slope = t_crit * std_err
```

In [120…

```
print('Baseline: ', baseline_enrollment)
print('2021-22: ', nationwide_trends['Total_Enrollment'][9])
print('Total Change %: ', total_change_pct)
print('Avg Change %: ', nationwide_trends['yoy_change'].mean())
print('Slope: ', slope)
print('Intercept: ', intercept)
print('95% CI: ', ci_slope)
print('R Squared: ', r_value**2)
print('P-Value: ', p_value)
```

```
Baseline:  17717229
2021-22:  15433062
Total Change %:  -12.892349023653754
Avg Change %:  -1.5150317578493226
Slope:  -1.2641234955277412
Intercept:  100.22380669942542
95% CI:  0.2766016400003222
R Squared:  0.9328116061637491
P-Value:  5.728672704293977e-06
```

P-Value is much less than the alpha value of 0.05, this means that the national enrollment trend is statistically significant.

Visualization

In [121…

```
## Enrollment Over time
px.line(nationwide_trends, x='Year', y='Total_Enrollment')
```

In [122…

```
## YoY Change Over time
px.line(nationwide_trends, x='Year', y='yoy_change')
```

In [123…

```
## Statistical Predictions
fig = go.Figure()

fig.add_trace(go.Scatter(
    x = nationwide_trends['Year'],
    y = nationwide_trends['enrollment_index'],
    mode = 'lines',
    name = 'Enrollment Index'
))

predictions = slope * years + intercept
fig.add_trace(go.Scatter(
    x=nationwide_trends['Year'],
    y=predictions,
    mode='lines',
    name=f'Linear Trend (R_squared={r_value**2:.3f})',
    line=dict(color='red', width=2, dash='dash')
))

fig.update_layout(
    title='Enrollment Index Over Time',
    xaxis_title='Year',
    yaxis_title='Enrollment Index',
    showlegend=True
)


fig.show()
```

Baseline is at 100. We see a clear downward trend of enrollment. The linear regression easily captures 93 % of the variance.

Question What is the national enrollment trend when aggregating across all states?

Answer There is a negative enrollment trend when aggregfating across all the states.

Question Which states have statistically significant declining trends?

In [124…

```
state_trends = state_enrollment.groupby(['State', 'Year']).agg({'Total_Enrol
state_trends
```

Out[124…

	State	Year	Total_Enrollment	Total_Pub_Under	Total_Priv_Under
0	Alabama	2012-13	265917	216535	49382
1	Alabama	2013-14	261188	213669	47519
2	Alabama	2014-15	259630	212458	47172
3	Alabama	2015-16	257650	212968	44682
4	Alabama	2016-17	258008	216115	41893
...	...	...	...	...	...
505	Wyoming	2017-18	30409	29945	464
506	Wyoming	2018-19	30058	30020	38
507	Wyoming	2019-20	29931	29678	253
508	Wyoming	2020-21	28456	27871	585
509	Wyoming	2021-22	27837	27095	742

510 rows × 5 columns

In [125…

```
states = state_enrollment['State'].unique()

total_change_pct_dict = {state: [] for state in states}
baseline_enrollment_dict = {state: [] for state in states}

for state in states:
    mask = state_trends['State'] == state
    df = state_trends[state_trends['State'] == state].copy()
    year_baseline = '2012-13'
    state_baseline = df[df['Year'] == year_baseline]['Total_Enrollment'].val
    baseline_enrollment_dict[state].append(state_baseline)
    total_change_pct_dict[state].append(((state_trends[state_trends['State']
    df['enrollment_index'] = (df['Total_Enrollment'] / state_baseline) * 100
    df['yoy_change'] = df['Total_Enrollment'].pct_change() * 100
    state_trends.loc[mask, 'enrollment_index'] = df['enrollment_index']
    state_trends.loc[mask, 'yoy_change'] = df['yoy_change']

state_trends.head()
```

Out[125…

	State	Year	Total_Enrollment	Total_Pub_Under	Total_Priv_Under	enrollment_i
0	Alabama	2012-13	265917	216535	49382	100.00
1	Alabama	2013-14	261188	213669	47519	98.22
2	Alabama	2014-15	259630	212458	47172	97.63
3	Alabama	2015-16	257650	212968	44682	96.89
4	Alabama	2016-17	258008	216115	41893	97.02

In [126…

```
avg_change_pct_dict = {state: [] for state in states}
slope_dict = {state: [] for state in states}
intercept_dict = {state: [] for state in states}
ci_slope_dict = {state: [] for state in states}
r_squared_dict = {state: [] for state in states}
p_value_dict = {state: [] for state in states}


state_years = np.arange(len(state_trends[state_trends['State'] == 'Alabama']

for state in states:
    slope,intercept, r_value, p_value, std_err = stats.linregress(state_year
    ci_slope = t.interval(0.95, len(state_years)-1, loc=slope, scale=std_err
    slope_dict[state].append(slope)
    intercept_dict[state].append(intercept)
    ci_slope_dict[state].append(ci_slope)
    r_squared_dict[state].append(r_value**2)
    p_value_dict[state].append(p_value)
    avg_change_pct_dict[state].append(state_trends[state_trends['State'] ==
```

In [127…

```
significant = []
insignificant = []
def print_state_info(state):
    for state in states:
        print("STATE: ", state)
        print('Baseline: ', baseline_enrollment_dict.get(state)[0])
        print('2021-22: ', state_trends[state_trends['State'] == state]['Tot
        print('Total Change %: ', total_change_pct_dict[state])
        print('Avg Change %: ', avg_change_pct_dict.get(state)[0])
        print('Slope: ', slope_dict.get(state)[0])
        print('Intercept: ', intercept_dict.get(state)[0])
        print('95% CI: ', ci_slope_dict.get(state)[0])
        print('R Squared: ', r_squared_dict.get(state)[0])
        print('P-Value: ', p_value_dict.get(state)[0])
        print('_______________________________________')
        print('')
for state in states:
    if p_value_dict.get(state)[0] < 0.05:
        significant.append(state)
```

```
    else:
        insignificant.append(state)
```

In [128…

```
print('Significant: ', significant)
print('Not Significant: ', insignificant)
```

```
Significant:  ['Alabama', 'Alaska', 'Arizona', 'Arkansas', 'Connecticut', 'D
istrict of Columbia', 'Florida', 'Hawaii', 'Idaho', 'Illinois', 'Indiana',
'Iowa', 'Kansas', 'Kentucky', 'Louisiana', 'Maine', 'Maryland', 'Massachuset
ts', 'Michigan', 'Minnesota', 'Mississippi', 'Missouri', 'Montana', 'Nebrask
a', 'New Hampshire', 'New Jersey', 'New Mexico', 'New York', 'North Carolin
a', 'North Dakota', 'Ohio', 'Oklahoma', 'Oregon', 'Pennsylvania', 'Rhode Isl
and', 'South Carolina', 'South Dakota', 'Tennessee', 'Utah', 'Vermont', 'Vir
ginia', 'Washington', 'West Virginia', 'Wisconsin', 'Wyoming']
Not Significant:  ['California', 'Colorado', 'Delaware', 'Georgia', 'Nevad
a', 'Texas']
```

The states that show not statistically significant declining trend are ['California', 'Colorado', 'Delaware', 'Georgia', 'Nevada', 'Texas']. This as to do with their p value being greater than 0.05.

In [129…

```
for state in significant:
    if slope_dict.get(state)[0] > 0:
        print(state + ': ', slope_dict.get(state)[0])
```

```
District of Columbia:  1.5156406938205302
Idaho:  1.6604377727889996
New Hampshire:  16.603854933943296
Utah:  5.219142158960169
```

Of the states with significant trends, only 4 have a positive trend, being D.C., Idaho, New Hampshire and Utah.

Visualization

In [130…

```
## Positive Enrollment Trend
fig = go.Figure()

for state in ['District of Columbia', 'Idaho', 'New Hampshire', 'Utah']:
    fig.add_trace(go.Scatter(x=state_trends[state_trends['State'] == state][

fig.update_layout(title='Enrollment Index Over Time', xaxis_title='Year', ya
fig.show()
```

In [131…

```
## Negative Enrollment Trend
fig = go.Figure()

for state in states:
    if state not in ['District of Columbia', 'Idaho', 'New Hampshire', 'Utah
        fig.add_trace(go.Scatter(x=state_trends[state_trends['State'] == sta

fig.update_layout(title='Enrollment Index Over Time', xaxis_title='Year', ya
fig.show()
```

In [132…

```
## Not Significant Trends
fig = go.Figure()

for state in insignificant:
    fig.add_trace(go.Scatter(x=state_trends[state_trends['State'] == state][

fig.update_layout(title='Enrollment Index Over Time', xaxis_title='Year', ya
fig.show()
```

Question which states have statistically significant declining trends?

Answer 'Alabama', 'Alaska', 'Arizona', 'Arkansas', 'Connecticut', 'Florida', 'Hawaii', 'Illinois', 'Indiana', 'Iowa', 'Kansas', 'Kentucky', 'Louisiana', 'Maine', 'Maryland', 'Massachusetts', 'Michigan', 'Minnesota', 'Mississippi', 'Missouri', 'Montana', 'Nebraska', 'New Jersey', 'New Mexico', 'New York', 'North Carolina', 'North Dakota', 'Ohio', 'Oklahoma', 'Oregon', 'Pennsylvania', 'Rhode Island', 'South Carolina', 'South Dakota', 'Tennessee', 'Vermont', 'Virginia', 'Washington', 'West Virginia', 'Wisconsin', 'Wyoming'

These states have statistically significant negative trends making them definitively decreasing for a cause other than randomness.

---

sub-question Which states show statistically significant growth trends?

In [133…

```
state_enrollment.head()
```

Out[133…

	State	Year	Total_Pub_Under	4y_Pub_Under	2y_Pub_Under	Total_Pub_Postba
0	Alabama	2012-13	216535	130260	86275	345
1	Alabama	2013-14	213669	129045	84624	346
2	Alabama	2014-15	212458	129916	82542	345
3	Alabama	2015-16	212968	131503	81465	344
4	Alabama	2016-17	216115	135080	81035	349

Using the Mann-Kendall test for this

In [134…

```
mk_results = []

for state in sorted(state_enrollment['State'].unique()):
    state_data = state_enrollment[state_enrollment['State'] == state].sort_v

    # Skip states with insufficient data
```

```
    if len(state_data) < 3:
        print(f"⚠ Skipping {state}: Insufficient data points ({len(state_dat
        continue

    values = state_data['Total_Enrollment'].values
    years_numeric = np.arange(len(values))

    # Mann-Kendall test
    mk_result = mk.original_test(values)

    # Linear regression for additional context
    slope, intercept, r_value, p_value_lr, std_err = stats.linregress(years_

    # Calculate percentage change metrics
    first_value = values[0]
    last_value = values[-1]
    total_change_pct = ((last_value - first_value) / first_value) * 100
    annual_change_pct = (slope / first_value) * 100

    # Determine if significant and direction
    is_significant = mk_result.p < 0.05
    trend_direction = mk_result.trend  # 'increasing', 'decreasing', or 'no

    mk_results.append({
        'State': state,
        'MK_Trend': trend_direction,
        'MK_P_Value': mk_result.p,
        'MK_Z_Score': mk_result.z,
        'MK_Tau': mk_result.Tau,
        'Significant_at_05': is_significant,
        'Total_Change_Pct': total_change_pct,
        'Annual_Change_Pct': annual_change_pct,
        'LR_Slope': slope,
        'LR_P_Value': p_value_lr,
        'R_Squared': r_value**2,
        'First_Year_Enrollment': first_value,
        'Last_Year_Enrollment': last_value,
        'N_Years': len(values)
    })

# Convert to DataFrame
mk_df = pd.DataFrame(mk_results)

print(f"\n✓ Analyzed {len(mk_df)} states")
print(f"✓ Mann-Kendall test completed for all states")
```

```
✓ Analyzed 51 states
✓ Mann-Kendall test completed for all states
```

In [135…

```
growth_states = mk_df[
    (mk_df['Significant_at_05'] == True) &
    (mk_df['MK_Trend'] == 'increasing')
].sort_values('Total_Change_Pct', ascending=False)

if len(growth_states) > 0:
    print(f"\n✓ {len(growth_states)} states show statistically significant G
```

```
    # Create nice display table
    display_cols = [
        'State',
        'Total_Change_Pct',
        'Annual_Change_Pct',
        'MK_P_Value',
        'R_Squared'
    ]

    print(growth_states[display_cols].to_string(index=False))

    # Summary statistics for growth states
    print(f"\n" + "-"*80)
    print("GROWTH STATES SUMMARY STATISTICS:")
    print("-"*80)
    print(f"Mean total change: {growth_states['Total_Change_Pct'].mean():.2f
    print(f"Median total change: {growth_states['Total_Change_Pct'].median()
    print(f"Range: {growth_states['Total_Change_Pct'].min():.2f}% to {growth
    print(f"Mean annual growth rate: {growth_states['Annual_Change_Pct'].mea

    # Strongest growth state
    strongest = growth_states.iloc[0]
    print(f"\nSTRONGEST GROWTH: {strongest['State']}")
    print(f"   Total change: {strongest['Total_Change_Pct']:.2f}%")
    print(f"   From {strongest['First_Year_Enrollment']:,.0f} to {strongest[
    print(f"   P-value: {strongest['MK_P_Value']:.4f}")

else:
    print("\n⚠ NO states show statistically significant growth at α = 0.05")
    print("\nLet's check for marginal growth (p < 0.10):")

    marginal_growth = mk_df[
        (mk_df['MK_P_Value'] < 0.10) &
        (mk_df['MK_Trend'] == 'increasing')
    ].sort_values('Total_Change_Pct', ascending=False)

    if len(marginal_growth) > 0:
        print(f"\n{len(marginal_growth)} states show marginally significant
        display_cols = ['State', 'Total_Change_Pct', 'MK_P_Value']
        print(marginal_growth[display_cols].to_string(index=False))
    else:
        print("No states show even marginal growth trends.")
```

```
✓ 4 states show statistically significant GROWTH

                State  Total_Change_Pct  Annual_Change_Pct  MK_P_Value  R_Squ
ared
        New Hampshire        144.707204          16.603855    0.000083   0.99
5941
                 Utah         37.106160           5.219142    0.000347   0.95
3748
                Idaho         17.320147           1.660438    0.020045   0.50
0452
District of Columbia         12.023313           1.515641    0.000677   0.88
1132

----------------------------------------------------------------------------
----
GROWTH STATES SUMMARY STATISTICS:
----------------------------------------------------------------------------
----
Mean total change: 52.79%
Median total change: 27.21%
Range: 12.02% to 144.71%
Mean annual growth rate: 6.25%

STRONGEST GROWTH: New Hampshire
   Total change: 144.71%
   From 66,770 to 163,391 students
   P-value: 0.0001
```

In [136…

```
trend_summary = mk_df.groupby(['MK_Trend', 'Significant_at_05']).size().rese

print("\nTrend Classification:")
for _, row in trend_summary.iterrows():
    sig_text = "Significant" if row['Significant_at_05'] else "Not Significa
    print(f"  {row['MK_Trend'].capitalize():12} ({sig_text:15}): {row['Count

# Statistical test: Are growth states significantly fewer than decline state
sig_growth = len(mk_df[(mk_df['Significant_at_05']) & (mk_df['MK_Trend'] == 
sig_decline = len(mk_df[(mk_df['Significant_at_05']) & (mk_df['MK_Trend'] ==

print(f"\nSignificant Growth vs Decline:")
print(f"  Growth:  {sig_growth} states")
print(f"  Decline: {sig_decline} states")
print(f"  Ratio:   {sig_decline}:{sig_growth}")

if sig_growth + sig_decline > 0:
    from scipy.stats import binomtest
    # Test if growth proportion is significantly less than 50%
    binom_result = binomtest(sig_growth, sig_growth + sig_decline, 0.5, alte
    print(f"  Binomial test p-value: {binom_result.pvalue:.4f}")
    print(f"  Conclusion: Growth states are {'significantly' if binom_result
```

```
Trend Classification:
  Decreasing   (Significant    ): 40 states
  Increasing   (Significant    ):  4 states
  No trend     (Not Significant):  7 states

Significant Growth vs Decline:
  Growth:  4 states
  Decline: 40 states
  Ratio:   40:4
  Binomial test p-value: 0.0000
  Conclusion: Growth states are significantly less common than decline state
s
```

In [137…

```
fig1 = go.Figure()

# Significant growth (green)
sig_growth_data = mk_df[(mk_df['Significant_at_05']) & (mk_df['MK_Trend'] ==
fig1.add_trace(go.Scatter(
    x=sig_growth_data['Annual_Change_Pct'],
    y=sig_growth_data['Total_Change_Pct'],
    mode='markers+text',
    name='Significant Growth',
    marker=dict(size=12, color='green', symbol='triangle-up'),
    text=sig_growth_data['State'],
    textposition='top center',
    hovertemplate='<b>%{text}</b><br>Annual: %{x:.2f}%<br>Total: %{y:.2f}%<e
))

# Significant decline (red)
sig_decline_data = mk_df[(mk_df['Significant_at_05']) & (mk_df['MK_Trend'] =
fig1.add_trace(go.Scatter(
    x=sig_decline_data['Annual_Change_Pct'],
    y=sig_decline_data['Total_Change_Pct'],
    mode='markers',
    name='Significant Decline',
    marker=dict(size=10, color='red', symbol='circle'),
    text=sig_decline_data['State'],
    hovertemplate='<b>%{text}</b><br>Annual: %{x:.2f}%<br>Total: %{y:.2f}%<e
))

# No significant trend (gray)
no_trend_data = mk_df[~mk_df['Significant_at_05']]
fig1.add_trace(go.Scatter(
    x=no_trend_data['Annual_Change_Pct'],
    y=no_trend_data['Total_Change_Pct'],
    mode='markers',
    name='Not Significant',
    marker=dict(size=8, color='lightgray', symbol='circle', opacity=0.5),
    text=no_trend_data['State'],
    hovertemplate='<b>%{text}</b><br>Annual: %{x:.2f}%<br>Total: %{y:.2f}%<e
))

fig1.add_hline(y=0, line_dash="dash", line_color="black", annotation_text="N
fig1.add_vline(x=0, line_dash="dash", line_color="black")

fig1.update_layout(
```

```
    title=f"State Enrollment Trends: Mann-Kendall Significance Test<br>" +
          f"<sub>{sig_growth} states growing significantly, {sig_decline} de
    xaxis_title="Average Annual Change (%)",
    yaxis_title="Total Change (%)",
    height=600,
    width=1000,
    showlegend=True,
    hovermode='closest'
)

fig1.show()

# Visualization 2: Distribution of p-values by trend direction
fig2 = make_subplots(
    rows=1, cols=2,
    subplot_titles=("Growth Trends", "Decline Trends"),
    specs=[[{"type": "histogram"}, {"type": "histogram"}]]
)

growth_trends = mk_df[mk_df['MK_Trend'] == 'increasing']
decline_trends = mk_df[mk_df['MK_Trend'] == 'decreasing']

fig2.add_trace(
    go.Histogram(
        x=growth_trends['MK_P_Value'],
        nbinsx=20,
        marker_color='green',
        name='Growth'
    ),
    row=1, col=1
)

fig2.add_trace(
    go.Histogram(
        x=decline_trends['MK_P_Value'],
        nbinsx=20,
        marker_color='red',
        name='Decline'
    ),
    row=1, col=2
)

# Add significance threshold line
fig2.add_vline(x=0.05, line_dash="dash", line_color="black", row=1, col=1)
fig2.add_vline(x=0.05, line_dash="dash", line_color="black", row=1, col=2,
               annotation_text="α=0.05")

fig2.update_xaxes(title_text="P-Value", row=1, col=1)
fig2.update_xaxes(title_text="P-Value", row=1, col=2)
fig2.update_yaxes(title_text="Number of States", row=1, col=1)

fig2.update_layout(
    title_text="Distribution of P-Values by Trend Direction",
    showlegend=False,
    height=400,
    width=1000
```

```
)

fig2.show()

# Visualization 3: If we have growth states, show their trends over time
if len(growth_states) > 0:
    fig3 = go.Figure()

    for _, state_row in growth_states.iterrows():
        state = state_row['State']
        state_data = state_enrollment[state_enrollment['State'] == state].sc

        fig3.add_trace(go.Scatter(
            x=state_data['Year'],
            y=state_data['Total_Enrollment'],
            mode='lines+markers',
            name=f"{state} ({state_row['Total_Change_Pct']:.1f}%)",
            hovertemplate='%{y:,.0f} students<extra></extra>'
        ))

    fig3.update_layout(
        title=f"Enrollment Trends: States with Significant Growth<br>" +
              f"<sub>Mann-Kendall test p < 0.05</sub>",
        xaxis_title="Academic Year",
        yaxis_title="Total Enrollment",
        height=500,
        width=1000,
        hovermode='x unified',
        showlegend=True
    )

    fig3.show()
```

Question: Which states show statistically growth trends in enrollment?

Answer: New Hampshire, Utah, Idaho, District of Columbia

---

sub-question What is the effect size of enrollment changes (Cohen's d)?

In [138…
```
total_en_year = state_enrollment.groupby(['Year']).agg({
    'Total_Enrollment': 'sum'
}).reset_index()
```

In [139…
```
total_en_year
```

Out[139…

	Year	Total_Enrollment
0	2012-13	17717229
1	2013-14	17459865
2	2014-15	17278075
3	2015-16	17021992
4	2016-17	16854168
5	2017-18	16745071
6	2018-19	16594731
7	2019-20	16549691
8	2020-21	15836385
9	2021-22	15433062

In [140…

```
std = total_en_year['Total_Enrollment'].std()
mean = total_en_year['Total_Enrollment'].mean()
ref = total_en_year['Total_Enrollment'][0]

cohen_d = np.abs((mean - ref) / std)
```

In [141…

```
print(cohen_d)
```

```
1.3790244595349892
```

This cohens d is approximately 1.38 which is a large effect size. Without the absolute value we know that the cohens d is negative which tells us that the effect size is quite large and negative meaning that it is structurally significant that the enrollment has decreased over the years.

Question: What is the effect size of enrollment changes (Cohen's d)?

Answer: 1.38

---

sub-question Are there inflection points where enrollment trends shifted direction?

We will utilize calculus to find inflection points.

In [142…

```
enrollment_trends = state_enrollment.groupby(['Year']).agg({
    'Total_Enrollment': 'sum'
}).reset_index()
```

In [143…

```
enrollment_trends['first_diff'] = enrollment_trends['Total_Enrollment'].diff
enrollment_trends['second_diff'] = enrollment_trends['first_diff'].diff()
enrollment_trends['third_diff'] = enrollment_trends['second_diff'].diff()
```

In [144…
```
enrollment_trends['infl'] = np.sign(enrollment_trends['third_diff'].fillna(0

infl_pts = enrollment_trends[enrollment_trends['infl']]
```

In [145…
```
infl_pts
```

Out[145…

	Year	Total_Enrollment	first_diff	second_diff	third_diff	infl
3	2015-16	17021992	-256083.0	-74293.0	-149867.0	True
4	2016-17	16854168	-167824.0	88259.0	162552.0	True
5	2017-18	16745071	-109097.0	58727.0	-29532.0	True
6	2018-19	16594731	-150340.0	-41243.0	-99970.0	True
7	2019-20	16549691	-45040.0	105300.0	146543.0	True
8	2020-21	15836385	-713306.0	-668266.0	-773566.0	True
9	2021-22	15433062	-403323.0	309983.0	978249.0	True

In [146…
```
fig = px.line(enrollment_trends, x='Year', y='Total_Enrollment')
fig.add_scatter(x=infl_pts['Year'], y=infl_pts['Total_Enrollment'], mode='ma
fig.update_layout(
    title="Enrollment Over Time",
    xaxis_title="Academic Year",
    yaxis_title="Total Enrollment",
    height=500,
    width=1000
)

fig.show()
```

Question: Are there inflection points where enrollment trends shifted direction?

Answer: Yes. The inflection points are in several areas most prominently howver is the downward shift in 2020 likely due to COVID-19.